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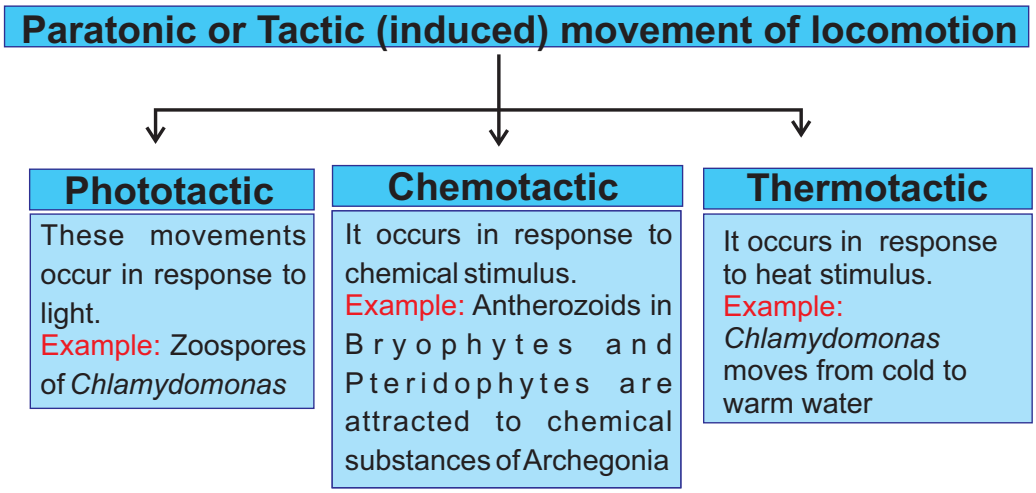


Figure 15.15: Types of Paratonic movements

A. Movements of locomotion

These movements include the movement of protoplasm inside the cell or movement of whole unicellular or multicellular plant body as in *Chlamydomonas*, gametes and zoospores.

i. Autonomic movements of locomotion

The movements arising from internal changes or internal stimuli of plant body is called **autonomic movements** of locomotion. This movement takes place due to the presence of cilia or flagella and movement of cytoplasm (Cyclosis).

ii. Paratonic or Tactic (induced) movements of locomotion

The movements due to external factors or stimuli like light, temperature and chemicals are called paratonic movement of locomotion (Figure 15.15).

B. Movement of curvature

In higher plants they are restricted only to bending or curvature of some of their parts. There are mainly two types:

They are i) Autonomic movement of curvature and ii) Paratonic movement of curvature.

i. Autonomic movements of curvature

The movement arising from internal changes or internal stimuli of plant body is called **autonomic movement of curvature**. This does not require any external stimulus. They are two types:

a. **Autonomic movement of growth:** It is of the following types:

1. **Hyponasty**: When growth is more on lower surface, petals show curvature on upper side and ultimately the flower becomes closed. Such type of movement is called **hyponasty.**

2. **Epinasty**: When the growth is more on upper surface, petals show curvature on the lower side and ultimately the flower opens. Such movement is called **epinasty.** The flower usually opens at high temperature and remains closed at low temperature (Figure 15.16).

3. **Nutation:** The growth of the stem apices occurs in a zig-zag manner. It is because the two sides of the stem apex alternatively grow more. Such growth movements are called as nutational movements. In some plants nutational movements allow the shoots apex to grow in helical path in upward